

Week 3 : Assignment 3

Detailed Solutions

Question 1

Sumit wants to contact one of his friends, but he remembers only the first 9 of the 10 digits of the contact number. He is sure that the last digit of the contact number is an odd number. He selects an odd number randomly. If the random variable X denotes the last digit of the contact number, calculate $\text{Var}(X)$.

- ☐ 5
- ☒ 8
- ☐ 33
- ☐ None of the above

Solution:

The possible odd digits are

$$X \in \{1, 3, 5, 7, 9\}.$$

Since one of the five odd digits is selected randomly, each value has probability

$$P(X = x) = \frac{1}{5}.$$

Therefore,

$$E(X) = \frac{1 + 3 + 5 + 7 + 9}{5}.$$

Hence,

$$E(X) = \frac{25}{5} = 5.$$

Now calculate $E(X^2)$:

$$E(X^2) = \frac{1^2 + 3^2 + 5^2 + 7^2 + 9^2}{5} = \frac{1 + 9 + 25 + 49 + 81}{5}.$$

$$E(X^2) = \frac{165}{5} = 33.$$

The variance is given by

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 33 - 5^2 = 33 - 25 = 8.$$

$\text{Var}(X) = 8$

Hence, the correct answer is 8.

Question 2

Suppose $X \sim N(\mu, 4)$. For $n = 20$ iid samples of X , the observed sample mean is $\bar{X} = 5.2$. What conclusion would a z -test reach if the null hypothesis assumes $\mu = 5$ against the alternative hypothesis $\mu \neq 5$ at significance level $\alpha = 0.05$? Use $F_Z^{-1}(0.025) = -1.9599$.

- ✓ Accept H_0
 - Reject H_0
-

Solution:

We have

$$X \sim N(\mu, 4).$$

Since the variance is 4, the standard deviation is

$$\sigma = \sqrt{4} = 2.$$

The hypotheses are

$$H_0 : \mu = 5 \quad \text{and} \quad H_1 : \mu \neq 5.$$

Since the alternative hypothesis contains $\neq$, this is a **two-tailed test**.

The z -test statistic is

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}.$$

Here,

$$\bar{X} = 5.2, \quad \mu_0 = 5, \quad \sigma = 2, \quad n = 20.$$

Therefore,

$$Z = \frac{5.2 - 5}{2/\sqrt{20}} = \frac{0.2}{2/\sqrt{20}}.$$

Since $\sqrt{20} \approx 4.4721$, we get $2/\sqrt{20} \approx 0.4472$.

Hence,

$$Z \approx \frac{0.2}{0.4472} \approx 0.4472.$$

For a two-tailed test at $\alpha = 0.05$, the critical values are

$$-1.9599 \quad \text{and} \quad 1.9599.$$

Thus, we reject H_0 only if $Z < -1.9599$ or $Z > 1.9599$. But

$$-1.9599 < 0.4472 < 1.9599.$$

Therefore, the test statistic does not fall in the rejection region.

Accept H_0

The correct option is **Accept H_0** .

Question 3

A box contains 8 items of which 2 are defective. A sample of 5 items is selected randomly without replacement. If the random variable X represents the number of defective items in the sample, find $E(X)$.

- ✓ 1.25
 - 5
-

Solution:

This is a **hypergeometric sampling** situation because:

- There are $N = 8$ total items.
- There are $K = 2$ defective items.
- We select $n = 5$ items.
- Sampling is without replacement.

For a hypergeometric random variable, the expected value is

$$E(X) = n \frac{K}{N}.$$

Substituting the values,

$$E(X) = 5 \left(\frac{2}{8} \right) = 5 \left(\frac{1}{4} \right) = \frac{5}{4} = 1.25.$$

$E(X) = 1.25$

The correct answer is **1.25**.

Question 4

Suppose $X \sim N(\mu, 9)$. For $n = 100$ iid samples of X , the observed sample mean is $\bar{X} = 11.8$. What conclusion would a z -test reach if the null hypothesis assumes $\mu = 10.5$ against an alternative hypothesis $\mu \neq 10.5$? Consider significance levels 0.10 and 0.05. (Select all that apply.)

- ☐ Accept H_0 at a significance level of 0.10.
- ☒ Reject H_0 at a significance level of 0.10.
- ☐ Accept H_0 at a significance level of 0.05.
- ☒ Reject H_0 at a significance level of 0.05.

Solution:

We have

$$X \sim N(\mu, 9) \quad \Rightarrow \quad \sigma = \sqrt{9} = 3.$$

The hypotheses are

$$H_0 : \mu = 10.5 \quad \text{and} \quad H_1 : \mu \neq 10.5.$$

Since the alternative hypothesis is $\mu \neq 10.5$, this is a two-tailed test. The z -statistic is

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{11.8 - 10.5}{3/\sqrt{100}} = \frac{1.3}{3/10} = \frac{1.3}{0.3} = 4.3333.$$

So the calculated test statistic is $Z \approx 4.33$.

At significance level $\alpha = 0.10$: For a two-tailed test, $\alpha/2 = 0.05$, giving critical values ± 1.645 . Since $4.33 > 1.645$, the test statistic lies in the rejection region.

Reject H_0 at $\alpha = 0.10$

At significance level $\alpha = 0.05$: For a two-tailed test, $\alpha/2 = 0.025$, giving critical values ± 1.96 . Since $4.33 > 1.96$, the test statistic again lies in the rejection region.

Reject H_0 at $\alpha = 0.05$

The correct choices are **Reject H_0 at significance level 0.10** and **Reject H_0 at significance level 0.05**.

Question 5

Let X and Y be two independent random variables with $\text{Var}(X) = 9$ and $\text{Var}(Y) = 3$. Find $\text{Var}(4X - 2Y + 6)$.

- ☐ 100
- ☐ 140

- ✓ 156
- None of the above

Solution:

We use the variance rule

$$\text{Var}(aX + bY + c) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y).$$

Since X and Y are independent, $\text{Cov}(X, Y) = 0$. Therefore,

$$\text{Var}(4X - 2Y + 6) = 4^2 \text{Var}(X) + (-2)^2 \text{Var}(Y).$$

The constant 6 does not affect variance. Thus,

$$= 16(9) + 4(3) = 144 + 12 = 156.$$

$\text{Var}(4X - 2Y + 6) = 156$

The correct option is **156**.

Question 6

The correlation coefficient of two random variables X and Y is 0.14. Their variances are given by 3 and 5. Compute $\text{Cov}(X, Y)$.

- -0.854
- 0.561
- ✓ 0.54
- None of the above

Solution:

The relationship between correlation and covariance is

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \quad \Rightarrow \quad \text{Cov}(X, Y) = \rho_{XY} \sigma_X \sigma_Y.$$

We are given $\rho_{XY} = 0.14$, $\text{Var}(X) = 3$, and $\text{Var}(Y) = 5$, so

$$\sigma_X = \sqrt{3}, \quad \sigma_Y = \sqrt{5}.$$

Hence,

$$\text{Cov}(X, Y) = 0.14(\sqrt{3})(\sqrt{5}) = 0.14\sqrt{15}.$$

Since $\sqrt{15} \approx 3.873$,

$$\text{Cov}(X, Y) \approx 0.14(3.873) \approx 0.5422.$$

Rounding to two decimal places,

$\text{Cov}(X, Y) \approx 0.54$

The correct option is **0.54**.

Question 7

When will you reject the null hypothesis?

- ☐ p value greater than α
- ☒ p value less than α
- ☐ p value equal to α
- ☐ None of the above

Solution:

In hypothesis testing, the p -value is compared with the significance level α . The decision rule is:

Reject H_0 if $p\text{-value} < \alpha$.

If $p\text{-value} > \alpha$, we do not reject H_0 .

The correct option is p **value less than** α .

Question 8

A sample of N observations is independently drawn from a normal distribution. The sample variance follows which distribution?

- ☐ Normal distribution
- ☐ Chi-square with N degrees of freedom
- ☒ Chi-square with $N - 1$ degrees of freedom
- ☐ t -distribution with $N - 1$ degrees of freedom

Solution:

Suppose $X_1, X_2, \dots, X_N$ are independent observations from a normal distribution $N(\mu, \sigma^2)$. The sample variance is

$$S^2 = \frac{1}{N-1} \sum_{i=1}^N (X_i - \bar{X})^2.$$

For a random sample from a normal population, the following result holds:

$$\frac{(N-1)S^2}{\sigma^2} \sim \chi_{N-1}^2.$$

Thus, the quantity involving the sample variance follows a chi-square distribution with $N - 1$ degrees of freedom.

Chi-square distribution with $N - 1$ degrees of freedom

The correct option is **Chi-square with $N - 1$ degrees of freedom.**

Question 9

A car manufacturer purchases car batteries from two different suppliers. Supplier X provides 55% of the batteries and Supplier Y provides the rest. 5% of all batteries from Supplier X are defective and 4% of all batteries from Supplier Y are defective. You select a battery from the bulk and find it to be defective. What is the probability that it is from Supplier X ?

- ☐ 0.0455
- ☐ 0.455
- ☒ 0.60
- ☐ 0.018

Solution:

We need to find $P(X | D)$, where X = battery is from Supplier X , Y = battery is from Supplier Y , and D = battery is defective.

We are given

$$P(X) = 0.55 \quad \Rightarrow \quad P(Y) = 1 - 0.55 = 0.45.$$

The probabilities of a defective battery are

$$P(D | X) = 0.05 \quad \text{and} \quad P(D | Y) = 0.04.$$

We use Bayes' theorem:

$$P(X | D) = \frac{P(D | X)P(X)}{P(D)}.$$

First calculate $P(D)$ using the law of total probability:

$$P(D) = P(D | X)P(X) + P(D | Y)P(Y) = (0.05)(0.55) + (0.04)(0.45).$$

Calculate each term: $(0.05)(0.55) = 0.0275$ and $(0.04)(0.45) = 0.018$.

Therefore,

$$P(D) = 0.0275 + 0.018 = 0.0455.$$

Now apply Bayes' theorem:

$$P(X | D) = \frac{(0.05)(0.55)}{0.0455} = \frac{0.0275}{0.0455} \approx 0.6044.$$

Rounding to two decimal places,

$$P(X | D) \approx 0.60$$

Therefore, the probability that the defective battery came from Supplier X is approximately **0.60**.

Question 10

Which one of the following is best measure of central tendency for categorical data?

- Mean
- Median
- ✓ Mode
- None of the above

Solution:

Categorical data consists of values that represent categories or labels (such as colors, brands, or types) rather than numerical quantities that can be meaningfully added, subtracted, or ordered on a numeric scale.

Mean requires numerical values that can be summed and divided, which is not meaningful for categories. **Median** requires the data to be ordered on a numeric or at least ordinal scale, which categorical (nominal) data does not generally support.

Mode, on the other hand, simply identifies the most frequently occurring category. Since it only relies on counting frequencies rather than performing arithmetic or ordering operations, it is well-defined for categorical data.

For example, if a survey records favourite fruit among {Apple, Banana, Apple, Mango, Apple}, the mode is “Apple,” since it occurs most frequently. Neither a mean nor a median can be meaningfully computed for this data.

Mode

The correct option is **Mode**.

Answer Summary

Question	Correct Answer
1	8
2	Accept H_0
3	1.25
4	Reject H_0 at 0.10 and 0.05
5	156
6	0.54
7	$p\text{-value} < \alpha$
8	χ^2 with $N - 1$ degrees of freedom
9	0.60
10	Mode